Int. J. Electron. Commun. (AEÜ) 164 (2023) 154619

Contents lists available at ScienceDirect

Int. J. Electron. Commun. (AEÜ)

journal homepage: www.elsevier.com/locate/aeue

Regular paper

Joint resource scheduling and trajectory design for multi-UAV-assisted uplink NOMA networks

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ARTICLE INFO

Keywords:
Unmanned aerial vehicle (UAV) communication
Non-orthogonal multiple access (NOMA)
Uplink transmission
Node clustering
Power control
Trajectory design

ABSTRACT

The combination of unmanned aerial vehicle (UAV) communication and non-orthogonal multiple access (NOMA) is of great significance to support massive connections in the Internet of things (IoT). This paper studies a multi-UAV-assisted communication network, where the UAVs are employed as aerial access points to receive data from IoT nodes with NOMA adopted for uplink transmission. The aim is to maximize the average sum rate of the network, with the requirements of the individual upload of each node. To handle the node clustering problem, the spectral clustering and maximum weight matching algorithm are first integrated to divide the IoT nodes. Then, the time-varying NOMA user grouping scheme is implemented accordingly. Furthermore, by considering the trajectory decomposition and successive hover-and-fly structure, the complex UAV trajectory problem is transformed into a linear programming and the optimal time-continuous trajectory is achieved. Extensive simulations validate the effectiveness of the proposed algorithms for the UAV-assisted NOMA network.

1. Introduction

1.1. Background and motivations

Unmanned aerial vehicles (UAVs) are expected to play an increasingly important role in civil, economic, military and even daily fields over recent years [1,2]. To satisfy the requirements of diversified applications, it is necessary to integrate UAVs and mobile communication techniques. Hence, UAV-assisted communication networks have attracted continuous research attentions and interests from industrial and academic communities. The benefits of UAV-assisted communications mainly rely on the controllable maneuverability, which enables UAVs to establish line-of-sight (LoS) links by constructing intelligent and controllable aerial platforms. To further elaborate the potential applications of UAVs in the next generation network, they can be leveraged as aerial base stations for alleviating communication pressure of terrestrial networks [3], as relays for providing emergency support in disaster areas [4], and as data-collectors for monitoring remote zones [5,6].

Despite the considerable advantages, the development of the UAV technology is restricted by the limitations of the battery energy and communication bandwidth. Therefore, improving the transmission efficiency of the UAV communication networks is emergent. It is believed that the non-orthogonal multiple access (NOMA) is one of the most essential and cost-effective technologies to compensate these restrictions, on account of its advantages of massive connections and high spectral efficiency [7]. In NOMA, the successive interference cancellation (SIC) is employed at the receivers, which allows UAVs serving different users simultaneously in a same resource block (RB) by multiplexing the users in other domains than frequency or time [8]. Since NOMA can support more connectivities in UAV networks, the resource scarcity problem is relieved and the UAVs are capable to complete various communication tasks within their endurance time. The integration of NOMA and UAVs has been widely proved to bring about predictable benefits, and can strike a good balance between system throughput and user fairness [9]. Three case studies are carried out in [10] to understand the UAV-assisted NOMA networks comprehensively from fundamental theory to actual implementation, but further research and analysis are needed with more practical scenarios.

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https://doi.org/10.1016/j.aeue.2023.154619
Received 11 January 2023; Accepted 7 March 2023
Available online 16 March 2023
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1.2. Related works

An overview on UAVs towards the fifth-generation and beyond networks has been summarized thoroughly in [11]. The existing work mainly focused on optimal deployment [10, 13], optimal trajectory [4, 11, 12, 14] and efficient resource allocation [12,13] to address different issues in UAV-assisted wireless networks. Specifically, the authors in [13] jointly optimized the UAV relay deployment, channel allocation, and relay assignment to maximize the communication capacity for a UAV-aided device-to-device network. To minimize the total energy consumption, the horizontal UAV trajectory and ground nodes' communication time were optimized in [14] with the derived closed-form propulsion power consumption model, while [6] optimized the three-dimensional (3D) trajectory and communication scheduling through iteratively updating the horizontal trajectory and vertical trajectory by block coordinate descent technique. To exploit expanded use of UAVs, more emerging concepts such as intelligent reflecting surface (IRS) [15], simultaneous wireless information and power transfer (SWIPT) [16,17], and integrated sensing and communication (ISAC) [18] are leveraged for ubiquitous connectivity and high reliability in both air-to-air and air-to-ground wireless communications. Specifically, the spectrum efficiency and the energy efficiency were maximized respectively in [19] for an IRS-UAV communication scheme, and the information and energy coverage of UAV-enabled SWIPT networks was investigated in [16] by considering both power splitting and time switching receiver architectures.

From the perspective of transmission technology, the existing research mainly focused on time division multiple access (TDMA) [11,19] or frequency division multiple access (FDMA) [12]. As mentioned previously, NOMA is inherently appealing to support UAV communication networks since it has the capability to improve the number of accessed users even for a large amount of occupied users, in comparison to the non-scalable orthogonal multiple access (OMA) schemes [20]. The fundamental design differences between uplink NOMA and downlink NOMA were detailedly analyzed in [21]. In [22], Ding et al. evaluated that NOMA has a better ergodic sum rate performance than OMA. Early effort mainly concentrated on UAV-assisted downlink NOMA networks, as the trajectory design and resource allocation were jointly optimized in [23], and the NOMA precoding was further investigated in [24]. Apart from maximizing system capacity and max–min rate, in [25], the total energy efficiency was maximized with quality of service (QoS) guarantee by considering the user scheduling, transmit power allocation, and trajectory optimization. Authors of [26] further investigated the reliability and security of the cognitive ambient backscatter communication network by invoking the outage probability and intercept probability with analytical derivation. More performance analysis metrics such as symbol error rate, bit error rate, and outage probability [27] are also studied previously. When extended to multi-UAV communication networks, a max–min optimization problem was formulated to achieve fair energy efficiency between UAVs in [28], where a simplified user pairing strategy based on relative distance was adopted. However, to the best of the authors' knowledge, rare works characterizes the joint optimization for uplink NOMA. The effect of the flight heights on system capacity was analyzed in single-UAV case [29] and multi-UAV case [30], respectively. Later in [31–33], the UAV deployment and user pairing problems were investigated in UAV-assisted uplink NOMA networks. In [34], the fundamental limits of rate/capacity regions were revealed by optimizing the one-dimensional UAV trajectory and resource allocation, and the optimality of the successive hover-and-fly structure was rigorously proved.

1.3. Original contributions and organization

By summarizing the related work, uplink NOMA has not been as much investigated for UAVs in the literature as the studies that examined downlink NOMA, where more key issues need to be handled in depth. In this paper, we focus on a multi-UAV-assisted uplink NOMA network, aiming to maximize the average sum rate through jointly considering the resource scheduling and horizontal trajectory design. It has been known that the K-means algorithm and matching theory are generally used for user clustering [31,32,35] and scheduling problems [30]. Hence, we first propose an improved novel algorithm based on the matching theory and spectral clustering to cluster the terrestrial nodes for UAVs. Then, on the basis of clustering results, we decompose the original problem into three sub-problems, namely dynamic user grouping, power control, and trajectory design to improve overall performance of the network. The main contributions of this work are listed as follows:

- First, we propose a multi-UAV-assisted uplink NOMA network, where the UAVs flies periodically to acquire data from massive IoT nodes, and the NOMA technology is applied to support several nodes access the same RB simultaneously to improve the spectral efficiency.
- To divide the massive terrestrial nodes into M clusters, we propose an improved spectral clustering based maximum weight matching (SCMWM) algorithm. This algorithm can satisfy the maximum service capacity constraint of each UAV, which is hard to be handled in conventional algorithms. Then, the optimization of time-varying NOMA user grouping can be implemented for IoT nodes to further enhance their upload capacity.
- Different from the generally used time discretization method for trajectory design, which has also been adopted in our previous work [36]. In this paper, we obtain the optimal time-continuous trajectory by trajectory decomposition, and accordingly transform the original complex problem into a linear programming one. During the flight, the UAV needs to properly design the communication scheduling and decoding order for the IoT nodes.
- Finally, extensive simulation results are presented to evaluate the performance of our proposed resource scheduling and trajectory design scheme. It is shown that the proposed SCMWM algorithm and the dynamic user grouping algorithm can both outperform the conventional clustering and grouping algorithms. Furthermore, the optimal trajectory design always achieves considerable rate gains over the compared two benchmark schemes with NOMA, FDMA, and TDMA protocols.

In the rest of this paper, Section 2 describes the system model of the UAV-assisted NOMA network and formulates the average sum rate maximization problem under data upload requirements. In Section 3, the joint optimization scheme of resource scheduling and trajectory design is proposed to solve the non-convex problem. Section 4 presents the simulation results to demonstrate the performance of our proposed scheme versus several benchmarks. Finally, Section 5 draws the conclusions.

Notations: Throughout the paper, matrices are denoted by bold upper-case letters (i.e., $\boldsymbol{A}$), column vectors are denoted by bold lower-case letters (i.e., $\boldsymbol{q}$), and scalars are denoted by normal letters (i.e. a). $\mathbb{E}(\cdot)$ denotes the statistical expectation, $|a|$ stands for the magnitude of a scalar, $\|\boldsymbol{q}\|$ represents the 2-norm of a vector, and $|\mathcal{A}|$ represents the cardinality of a set.

2. System model

2.1. System description

We assume a network as depicted in Fig. 1, where K terrestrial nodes are distributed randomly in the given region, with their individual data upload requirements within duration T, and M UAVs are deployed as aerial access points to collect data from the nodes. Such a UAV-assisted network architecture can be applied to many delay-insensitive IoT systems such as smart farming, forest monitoring, rare

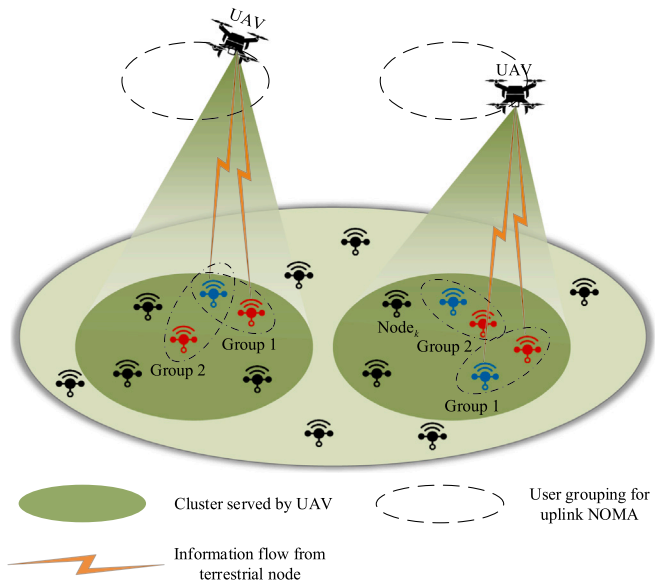

Fig. 1. The architecture of the multi-UAV-assisted uplink NOMA network.

animal monitoring, traffic supervision, etc [30]. The sets of the UAVs and IoT nodes are denoted as $\mathcal{M} = \{1, \cdots, M\}$ and $\mathcal{K} = \{1, \cdots, K\}$, respectively. The mth UAV flies at the altitude of H and the radius of e_m over a finite communication period $\mathcal{T} \triangleq (0, T]$, to construct its circular coverage area. Several IoT nodes can upload data to one UAV simultaneously via NOMA technology over a single RB to improve the spectral efficiency,[1] and different UAVs can provide service using different orthogonal RBs. We assume that all IoT nodes must be covered by the M UAVs, and let the set of the IoT nodes served by the mth UAV be denoted by $\mathcal{A}_m$, and thus we have $\sum_{m\in\mathcal{M}} |\mathcal{A}_m| = K$, $\bigcup_{m\in\mathcal{M}} \mathcal{A}_m = \mathcal{K}$ and $\mathcal{A}_{m_1} \bigcap \mathcal{A}_{m_2} = \emptyset$ $(\forall m_1, m_2 \in \mathcal{M},\ m_1 \neq m_2)$.

Without loss of generality, a 3-D Cartesian coordinate system is adopted, in which the location of the kth IoT node is denoted as $\boldsymbol{g}_k$, and the time-varying horizontal coordinate of the mth UAV is denoted as $\boldsymbol{q}_m(t)$, with its center of the circular trajectory being $(\boldsymbol{u}_m{}^{\mathrm{T}}, H)$. At time instant $t \in \mathcal{T}$, the angle of the UAVs counterclockwise from the x axis is denoted by $\boldsymbol{\theta}(t) = \left[\theta_1(t), \ldots, \theta_M(t)\right]^{\mathrm{T}}$ in radian, thus $\boldsymbol{q}_m(t) = \boldsymbol{u}_m + \left[e_m \cos\left(\theta_m(t)\right), e_m \sin\left(\theta_m(t)\right)\right]^{\mathrm{T}}$. The distance between the UAV and the kth node can be calculated by $D_{k,m}(t) = \left\|\left(\boldsymbol{u}_m - \boldsymbol{g}_k + [e_m \cos\left(\theta_m(t)\right), e_m \sin\left(\theta_m(t)\right)]; H\right)\right\|$. Considering the anti-collision process in the UAV system, their trajectories $\boldsymbol{q}_m(t)$ should satisfy

$$\left\|\boldsymbol{q}_{m1}(t) - \boldsymbol{q}_{m2}(t)\right\| \leq d_{\min}, \tag{1}$$

where $m_1, m_2 \in \mathcal{M}$ and $m_1 \neq m_2$. The channel coefficient between the mth UAV and the kth ground node at time instant t is modeled as [37]

$$h_{k,m}(t) = \sqrt{\frac{P_{k,m}^{\mathrm{LoS}}(t)\beta_0}{D_{k,m}^{\tilde{\alpha}}(t)} + \frac{\xi P_{k,m}^{\mathrm{NLoS}}(t)\beta_0}{D_{k,m}^{\tilde{\alpha}}(t)}}\,\tilde{h}_{k,m}(t), \tag{2}$$

where $P_{k,m}^{\mathrm{LoS}}(t)$ and $P_{k,m}^{\mathrm{NLoS}}(t)$ are the probabilities of LoS link and non-LoS (NLoS) link, respectively. β_0 indicates the path loss at reference distance of $d_0 = 1$ m, $\tilde{\alpha}$ is the path loss exponent, ξ is the additional propagation loss of the NLoS link, and $\tilde{h}_{k,m}(t)$ is a complex-valued random variable with $\mathbb{E}\left(\left|\tilde{h}_{k,m}(t)\right|^2\right) = 1$.

The bandwidth of one RB allocated to each UAV-enabled subnetwork is given by B Hz, and $T_s = 1/B$ denotes the corresponding symbol duration. Let $V_{\max} \geq 0$ denote the maximum UAV flight speed, then we have $w_{\max}^m = V_{\max}/e_m$, where $w_{\max}^m$ is the maximum angular velocity of the mth UAV. As within each symbol interval, the UAV's location change can be considered little as compared to the flight altitude, then the influence of Doppler frequency shift on channel coefficient can be considered negligible [38]. The LoS link probability is generally dependent on the propagation environment and elevation angle [37] between UAVs and users, and the free-space path loss model with $P_{k,m}^{\mathrm{LoS}}(t) = 1$ is widely adopted in UAV trajectory design by leveraging the mobility of UAVs (see, e.g., [39,40]). Therefore, in this work we focus on the average propagation loss and ignore the small-scale fluctuations which is mainly caused by multipath fading and Doppler frequency shift [34].

At time instant t, we define an indicator matrix $\Lambda(t) \in \mathbb{R}^{K\times M}$ to represent the communication scheduling of UAVs with IoT nodes. The element at the kth row and the mth column $\lambda_{k,m}(t) \in \{0, 1, 2\}$: $\lambda_{k,m}(t) > 0$ indicates that the kth node is served by the mth UAV, otherwise, $\lambda_{k,m}(t) = 0$. Besides, $\lambda_{k,m}(t) = 1$ or 2 represents that the kth node is ordered as node 1 or node 2 in its NOMA group.[2] It is assumed that each UAV can only serve one NOMA group and each group can only upload data to one UAV at one time instant.

Let $s_k(t)$ denote the signal transmitted by the kth IoT node to the mth UAV. Accordingly, the received signal at each UAV is

$$y_m(t) = \sum_{k\in\mathcal{A}_m} \sqrt{h_{k,m}(t)}\, s_k(t) + n(t), \tag{3}$$

where $n(t)$ is the complex additive white Gaussian noise (AWGN) with zero mean and variance σ^2. Define $\boldsymbol{p}(t) = [P_1(t), \ldots, P_K(t)]^{\mathrm{T}}$ as the power control vector with $P_k(t) = \mathbb{E}\left(\left|s_k(t)\right|^2\right)$, and the power constraint of each node can be given as

$$0 \leq P_k(t) \leq P_{\max}, \quad \forall k \in \mathcal{K},\ t \in \mathcal{T} \tag{4}$$

where $P_{\max}$ is the maximum uplink transmission power constraint for each node.

2.2. Problem formulation

Once the signals from IoT nodes are received at the UAV, UAV receiver can adopt SIC technology to eliminate the interference generated by other nodes sharing the same RB. For downlink NOMA, the receiver first detects the user with the worst channel condition, while in the uplink transmission the decoding order is reversed. Let vector $\boldsymbol{\pi}_m(t) = \left[\pi_m^1(t), \ldots, \pi_m^N(t)\right]$ denotes the node index to indicate decoding order, where $h_{\pi_m^N(t),m}(t) \geq \cdots \geq h_{\pi_m^1(t),m}(t)$. Specifically, it indicates that the UAV receiver first decodes the node $\pi_m^1(t)$'s message in downlink network. For uplink network in this paper, the node $\pi_m^N(t)$'s message is first detected, then the interference from $s_{\pi_m^N(t)}(t)$ is canceled for decoding $s_{\pi_m^{N-1}(t)}(t)$, and followed so on until $s_{\pi_m^1(t)}(t)$. Employing the user index vector $\boldsymbol{\pi}_m(t)$, the necessary power constraint for efficient SIC in two-user NOMA group can be expressed as:

$$P_{\pi_m^2(t)}(t) h_{\pi_m^2(t),m}(t) - P_{\pi_m^1(t)}(t) h_{\pi_m^1(t),m}(t) \geq P_{\mathrm{tol}}\ . \tag{5}$$

Then the achievable rate (bps/Hz) of the kth node can be given by

$$r_{k,m}(t) = \log_2\left(\frac{\sigma^2 + \sum_{n=1}^{\lambda_{k,m}(t)} P_k(t) h_{\pi_m^n(t),m}(t)}{\sigma^2 + \sum_{n=1}^{\lambda_{k,m}(t)-1} P_k(t) h_{\pi_m^n(t),m}(t)}\right). \tag{6}$$

[1] Considering the battery and volume limitation of UAVs and the complexity of SIC at the receiver, an example with two users in each NOMA group is adopted in this paper, similarly as in [28]. Our optimization principles can be easily extended to the cases with more nodes in a group, for which the grouping strategy should be modified.

[2] Without loss of generality, it is assumed that the nodes in each NOMA group are reordered according to their channel gains. I.e., denoting the channel coefficient of node n by h_n, if $m > n$, generally we have $h_m > h_n$.

Under the given trajectory $\boldsymbol{q}_m(t)$, the average sum rate of all nodes in $\mathcal{A}_m$ within duration T is expressed as

$$R_m = \frac{1}{T}\int_0^T \left(\sum_{k\in\mathcal{A}_m} r_{k,m}(t)\right)\mathrm{d}t. \tag{7}$$

The objective is to improve the spectral efficiency of all IoT nodes in this UAV-assisted network. Hence, the average sum rate maximization problem is formulated as

$$(\mathbf{P1}): \max_{\boldsymbol{A}(t),\boldsymbol{p}(t),\{\boldsymbol{q}_m(t)\}} \sum_{m\in\mathcal{M}} R_m \tag{8}$$

$$\text{s.t.} \quad \sum_{m\in\mathcal{M}}\int_0^T r_{k,m}(t)\mathrm{d}t \geq \eta_c,\ \forall k\in\mathcal{K} \tag{8a}$$

$$0 \leq |\dot{\boldsymbol{q}}_m(t)| \leq V_{\max},\ \forall t\in\mathcal{T} \tag{8b}$$

$$\boldsymbol{q}_m(0) = \boldsymbol{q}_m(T),\ \forall m\in\mathcal{M} \tag{8c}$$

$$|\mathcal{A}_m| \leq K_{\max},\ \forall m\in\mathcal{M} \tag{8d}$$

$$(1),\ (4),\ (5),$$

where constraint (8a) represents the data upload requirements of the nodes, constraint (8b) is the limitation on UAVs' maximum velocity, and constraint (8c) ensures that the UAVs can return to the original positions and restart the next flight period to serve ground nodes. Considering the restrained resources of each UAV-assisted subnetwork and the user fairness among M clusters, constraint (8d) provides the maximum service capacity limitation of each UAV. Obviously, the problem (**P1**) is non-convex and constitutes infinite number of optimization variables that are coupled in continuous functions over time, which is highly challenging to be optimally solved.

3. Proposed solution

3.1. Nodes clustering and dynamic user grouping

As can be readily seen from (8), the clustering results of the terrestrial nodes should first be determined before solving the user grouping and power control problems. To ensure the user fairness, it has been assumed that each UAV can serve up to $K_{\max}$ users at one time instant. Since the locations of all IoT nodes are known, we can cluster the nodes based on their location proximity to shorten their communication distance. Hence, we propose an improved SCMWM algorithm to divide the nodes into M clusters. First, the spectral clustering algorithm is applied to achieve M initial clusters, which has been verified to outperform the traditional K-means algorithm [41]. Specifically, the pairwise similarities of the randomly distributed node locations $\{\boldsymbol{g}_1,\ldots,\boldsymbol{g}_K\}$ are measured by $s_{ij} = s(\boldsymbol{g}_i,\boldsymbol{g}_j)$, and the similarity function is generally based on Gaussian kernel function given by

$$s_{ij} = \exp\left(-\frac{\left\|\boldsymbol{g}_i-\boldsymbol{g}_j\right\|^2}{2\sigma_d^2}\right), \tag{9}$$

where σ_d denotes the neighborhood coefficient, and the corresponding similarity matrix can be denoted by $\boldsymbol{S} = (s_{ij})_{i,j=1,\ldots,K}$. Then the conventional spectral clustering algorithm can be utilized to form clusters of the IoT nodes based on their Gaussian distance similarity matrix, which cannot handle the maximum service capacity constraint. To address this constraint, by calculating the centroid $\boldsymbol{u}_m$ of the nodes in the mth initial cluster, we improve the initial clustering results by the maximum weight matching theory. Define the distance relationship between $\boldsymbol{g}_k$ and $\boldsymbol{u}_m$ as a weighted bipartite graph $G = (V,E)$, where $V = (A,B)$ with $A = \{a_1,\ldots,a_M\}$ and $B = \{b_1,\ldots,b_K\}$ represents the vertices, and $E = \{E_{k,m}(\forall k\in B, m\in A)\}$ with the weights $\omega_{k,m} = -\|\mathbf{g}_k-\mathbf{u}_m\|$ represents the edges. To handle the constraint (8d), we construct a new weighted bipartite graph $G' = (V',E')$ by expanding the UAV set to $K_{\max}$ pieces (pcs), where $V' = (A',B)$, and $A' = \left\{a_1^1,\ldots,a_1^{K_{\max}},\ldots,a_M^1,\ldots,a_M^{K_{\max}}\right\}$, as shown in Fig. 2. Then the clustering problem is equivalent to

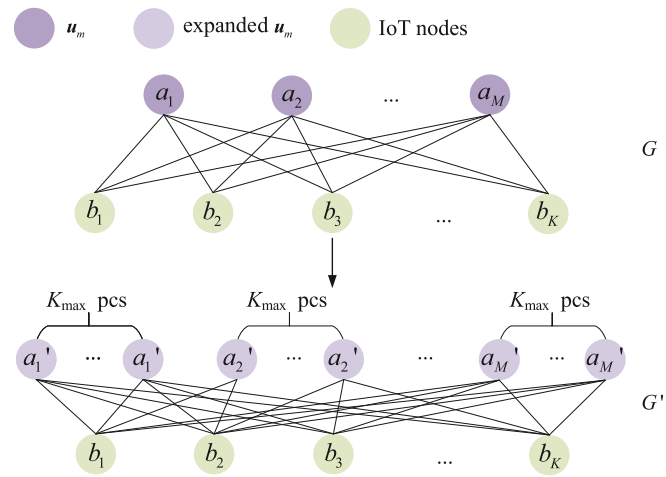

Fig. 2. The initial and expanded weighted bipartite graphs.

a standard maximum weight matching problem, and can be easily solved by Kuhn–Munkras algorithm. The maximum weight matching algorithm is then repeated by updating $\boldsymbol{u}_m$ and G', until the clustering results are unchanged. The improved SCMWM algorithm can make nodes distributed in each cluster relatively uniformly, which is specified in Algorithm 1.

Algorithm 1 The Improved SCMWM Clustering Algorithm

Require: M, K, $K_{\max}$, $\{\boldsymbol{g}_k\}$
1: **Init**: Iteration index $n = 0$.
2: Compute the Gaussian distance similarity matrix $\boldsymbol{S}$ using formula (9).
3: Achieve the initial clusters $\left\{\mathcal{A}_m^{(n)}\right\}$ by spectral clustering algorithm in [42], and calculate the centroids $\left\{\boldsymbol{u}_m^{(n)}\right\}$.
4: **Repeat**:
5: Generate a weighted bipartite graph $G = (V,E)$ by $\left\{\boldsymbol{u}_m^{(n)}\right\}$ and $\boldsymbol{g}_k$, and expand G to G' according to the maximum service capacity of each UAV;
6: Update the optimal clustering results $\left\{\mathcal{A}_m^{(n+1)}\right\}$ via Kuhn–Munkras algorithm;
7: Update $\left\{\boldsymbol{u}_m^{(n+1)}\right\}$ according to $\left\{\mathcal{A}_m^{(n+1)}\right\}$.
8: $n = n+1$;
9: **Until** $\left\{\mathcal{A}_m^{(n)}\right\} = \left\{\mathcal{A}_m^{(n-1)}\right\}$.
Output: $\{\mathcal{A}_m\}$

As mentioned above, two users are paired into one NOMA group as an example to analyze the grouping principles efficiently. The conventional user grouping scheme is usually fixed, which first classifies the users into high channel gain users and low channel gain users, and then distributes the high channel gain users into different NOMA groups, as they can significantly contribute to the throughput of a group. However, this fixed grouping scheme improves the high channel gain users' throughput at the cost of low channel gain users' performance. Leveraging the mobility of UAVs, they can communicate with the terrestrial nodes in turns, and thus a low-complexity and time-varying grouping scheme can be proposed. In this time-varying scheme, each node can experience a high channel gain with the closer UAV, and the highest channel gain user is paired with the second highest channel gain user into a NOMA group at each time instant t. The optimized trajectories of the UAVs eliminate the need for nodes to be divided into high and low channel gain users, because the channel gain of each user can be improved during a flight period.

To achieve the optimal user grouping solutions, it generally requires the exhaustive method to consider all possible combinations of users [43]. Obviously, the computational complexity is not affordable for practical UAV networks, especially for time-varying user grouping scheme. As such, we resort to a low-complexity grouping scheme in two steps to obtain a sub-optimal solution.

Based on the achieved clustering results $\mathcal{A}_m$, we reestablish M coordinate systems with $\boldsymbol{u}_m$ as the origin. Meanwhile, the nodes in $\mathcal{A}_m$ are reordered as $\mathcal{A}_m{}' = \{1, \dots, K_m\}$, and the angle of the ith node in $\mathcal{A}_m{}'$ counterclockwise from the x axis is denoted by $\phi_{m,i}$. Actually, the nodes in $\mathcal{A}_m{}'$ are ordered according to $\phi_{m,1} < \cdots < \phi_{m,K_m}$. The coordinate of the ith node in $\mathcal{A}_m{}'$ is denoted by $\boldsymbol{g}_{m,i}{}'$, and its distance to the UAV is given by $d_{m,i}{}'(t)$, while $\rho_m(i)$ denotes its initial index in $\mathcal{A}_m$. The flight radius of the mth UAV can be calculated as $Z_m = \left(\sum_{i\in\mathcal{A}_m{}'} |\boldsymbol{g}_{m,i}{}'|\right)/K_m$. To satisfy the anti-collision constraint, if $|\mathbf{u}_{m1} - \mathbf{u}_{m2}| < d_{\min} + Z_{m1} + Z_{m2}$ ($m_1, m_2 \in \mathcal{M}$ and $m_1 \neq m_2$), the radius will be updated as

$$Z_{m1(m2)} = \left(\sum_{i\in\mathcal{A}_{m1(m2)}{}'} \left|\boldsymbol{g}_{m1(m2),i}{}'\right|\right) \times \frac{|\boldsymbol{u}_{m1} - \boldsymbol{u}_{m2}| - d_{\min}}{\sum_{i\in\mathcal{A}_{m1}{}'} |\boldsymbol{g}_{m1,i}{}'| + \sum_{i\in\mathcal{A}_{m2}{}'} |\boldsymbol{g}_{m2,i}{}'|}. \tag{10}$$

Step 1: To find the nearest node to the mth UAV without exhaustive search at time instant t, we first connect nodes $\boldsymbol{g}_{m,i}{}'$ and $\boldsymbol{g}_{m,i+1}{}'$ ($i \in \mathcal{A}_m{}', i \neq K_m$),[3] and make the perpendicular bisector of this line intersect with the UAV trajectory. If the intersection exists, define its angle counterclockwise from x axis as $\Psi_{m,i}$, then it can be easily achieved that

$$\lambda_{\rho_m(i),m}(t) = 2,\ \Psi_{m,(i-1)} \le \theta_m(t) < \Psi_{m,i} \tag{11}$$

where $\Psi_{m,0} = \Psi_{m,K_m} - 2\pi$. However, if the ith node is far away from the UAV, there may be no intersection between the perpendicular bisector of its connection line with other nodes and the UAV trajectory. In this case, the UAV may not provide service to this node according to the selection criterion, thus in this case we make $\Psi_{m,i} = \left(\phi_{m,i} + \phi_{m,(i+1)}\right)/2$ to ensure the fairness of all nodes.

The above analysis is based on the assumption that $\Psi_{m,(i+1)} \ge \Psi_{m,i}$, and we have the following lemma with $\Psi_{m,(i+1)} < \Psi_{m,i}$.

Lemma 1. *For given $\{\Psi_{m,i}\}$, if $\Psi_{m,(i+1)} < \Psi_{m,i}$, we have*

$$\lambda_{\rho_m(i),m}(t) = 2, \quad \Psi_{m,(i-1)} \le \theta_m(t) < \Psi_{m,(i+1)} \tag{12a}$$

$$\lambda_{\rho_m(i+2),m}(t) = 2, \quad \Psi_{m,i} \le \theta_m(t) < \Psi_{m,(i+2)} \tag{12b}$$

Proof. Please refer to Appendix. ■

Based on Lemma 1, it is evident that when $\Psi_{m,(i+1)} < \Psi_{m,i}$, the distance from the UAV to the $(i+1)$-th node is always farther than to the ith and $(i+2)$-th node. Therefore, we let $\lambda_{\rho_m(i+1),m}(t) = 2$ when $\Psi_{m,(i+1)} \le \theta_m(t) < \Psi_{m,i}$ to ensure the $(i+1)$-th node be serviced. According to the above analysis, we can clarify the solution to select the first node into the NOMA group as follows. The initially obtained series $\{\Psi_{m,i}\}$ should first be re-sorted in ascending order, which is updated as $\{\tilde{\Psi}_{m,i}\}$. Then the communication scheduling variable at time instant t can be given as

$$\lambda_{\rho_m(i),m}(t) = 2,\ \tilde{\Psi}_{m,(i-1)} \le \theta_m(t) < \tilde{\Psi}_{m,i} \tag{13}$$

where $\tilde{\Psi}_{m,0} = \tilde{\Psi}_{m,K_m} - 2\pi$. According to $\{\tilde{\Psi}_{m,i}\}$ and (13), nodes in $\mathcal{A}_m{}'$ can be sequentially distributed into a NOMA group and be served by the UAV during the whole flight period. Specifically, when $\tilde{\Psi}_{m,(i-1)} \le \theta_m(t) < \tilde{\Psi}_{m,i}$, the ith node in $\mathcal{A}_m{}'$ is first included as node 2 in the NOMA group.

[3] if $i = K_m$, it will be connected with $\boldsymbol{g}_{m,1}{}'$.

Step 2: Then at each time instant, the second nearest node to the UAV should be paired with the previously selected node to complete the group formation. Suppose that there exists $\Gamma_{m,i}$ nodes to be paired with the ith node sequentially within $\left[\tilde{\Psi}_{m,(i-1)}, \tilde{\Psi}_{m,i}\right)$. The communication scheduling variable can be further achieved as

$$\lambda_{f_{m,i}(j),m}(t) = 1,\ \psi_{m,i,(j-1)} \le \theta_m(t) < \psi_{m,i,j} \tag{14}$$

where $j \in \mathcal{Z}_{m,i} = \{1, \dots, \Gamma_{m,i}\}$. The function $f_{m,i}(j)$ denotes the jth paired node's initial index in $\mathcal{A}_m$ within $\left[\tilde{\Psi}_{m,(i-1)}, \tilde{\Psi}_{m,i}\right)$, and it is paired with the ith node within $\left[\psi_{m,i,(j-1)}, \psi_{m,i,j}\right)$. Besides, there exists $\psi_{m,i,0} = \tilde{\Psi}_{m,(i-1)}$ and $\psi_{m,i,\Gamma_{m,i}} = \tilde{\Psi}_{m,i}$. The above analysis still entails an exhaustive search method to find the nearest one among the remaining nodes, and is challenging to implement due to the continuous time and large amount of nodes. Consequently, in our proposed grouping scheme, only the two nodes adjacent to the ith node are compared and selected, and the sub-optimal communication scheduling variable can be obtained similarly as in step 1. Obviously, the optimal $\Lambda^*(t)$ can not only indicate which two nodes are scheduled to upload data to the UAV via NOMA, but also present the decoding order of the users. Then $\pi_m^{n^*}(t)$ can be achieved by finding the index of the element with value n in the mth column of $\Lambda^*(t)$. The sub-optimal low-complexity grouping scheme is described comprehensively in Algorithm 2.

Algorithm 2 The Dynamic User Grouping Algorithm

Require: $\{\mathcal{A}_m\}$
Reestablish M coordinate systems and update $\mathcal{A}_m{}'$, $\phi_{m,i}$, $\{\boldsymbol{g}_{m,i}{}'\}$.
for m=1 to M **do**
  for i=1 to K_m **do**
    Connect $\boldsymbol{g}_{m,i}{}'$ and $\boldsymbol{g}_{m,(i+1)}{}'$ ($\boldsymbol{g}_{m,(K_m+1)}{}' = \boldsymbol{g}_{m,1}{}'$), and let its perpendicular bisector intersect with the UAV trajectory.
    if the intersection exists **then**
      Denote the angle of the intersection as $\Psi_{m,i}$.
    else $\Psi_{m,i} = \left(\phi_{m,i} + \phi_{m,(i+1)}\right)/2$.
    end if
  end for
end for
Re-sort $\{\Psi_{m,i}\}$ as $\{\tilde{\Psi}_{m,i}\}$, and update the communication scheduling variables $\lambda_{k,m}(t)$ by (13).
for m=1 to M **do**
  for i=1 to K_m **do**
    Connect $\boldsymbol{g}_{m,(i-1)}{}'$ and $\boldsymbol{g}_{m,(i+1)}{}'$ ($\boldsymbol{g}_{m,0}{}' = \boldsymbol{g}_{m,K_m}{}'$), and let its perpendicular bisector intersect with the UAV trajectory.
    if the intersection exists **then**
      Denote the angle of the intersection as $\psi_{m,i,1}$.
      if $\tilde{\Psi}_{m,(i-1)} < \psi_{m,i,1} < \tilde{\Psi}_{m,i}$ **then**
        $\lambda_{\rho_m(i-1),m}(t) = 1, \theta_m(t) \in \left(\tilde{\Psi}_{m,(i-1)}, \psi_{m,i,1}\right)$.
        $\lambda_{\rho_m(i+1),m}(t) = 1, \theta_m(t) \in \left(\psi_{m,i,1}, \tilde{\Psi}_{m,i}\right)$.
      elseif $\psi_{m,i,1} \le \tilde{\Psi}_{m,(i-1)}$ **then**
        $\lambda_{\rho_m(i+1),m}(t) = 1, \theta_m(t) \in \left(\tilde{\Psi}_{m,(i-1)}, \tilde{\Psi}_{m,i}\right)$.
      elseif $\psi_{m,i,1} \ge \tilde{\Psi}_{m,i}$ **then**
        $\lambda_{\rho_m(i-1),m}(t) = 1, \theta_m(t) \in \left(\tilde{\Psi}_{m,(i-1)}, \tilde{\Psi}_{m,i}\right)$.
      end if
    else
      if $d_{m,(i-1)}{}'(\Psi_{m,i}) \le d_{m,(i+1)}{}'(\Psi_{m,i})$ **then**
        $\lambda_{\rho_m(i-1),m}(t) = 1, \theta_m(t) \in \left(\tilde{\Psi}_{m,(i-1)}, \tilde{\Psi}_{m,i}\right)$.
      elseif $d_{m,(i+1)}{}'(\Psi_{m,i}) < d_{m,(i-1)}{}'(\Psi_{m,i})$ **then**
        $\lambda_{\rho_m(i+1),m}(t) = 1, \theta_m(t) \in \left(\tilde{\Psi}_{m,(i-1)}, \tilde{\Psi}_{m,i}\right)$.
      end if
    end if
  end for
end for
Output: $\Lambda^*(t)$

Based on the given nodes clustering and user grouping results,[4] the power and trajectory optimization can be individually performed in each UAV-enabled subnetwork. The primal problem can be decomposed into M sub-problems, and the mth sub-problem is formulated as

$$(\mathbf{P2}): \max_{p(t),q_m(t)} R_m \tag{15}$$

$$\text{s.t.} \quad \int_0^T r_{k,m}(t)\mathrm{d}t \geq \eta_c, \forall k \in \mathcal{A}_m \tag{15a}$$

(4), (5), (8b), (8c).

3.2. Closed-form solution for power control

It can be observed from (13) and (14) that $\lambda_{k,m}{}^*(t)$ remains unchanged when $\psi_{m,i,(j-1)} \leq \theta_m(t) < \psi_{m,i,j}$, which indicates the fixed user grouping results and decoding order. During the whole flight period of the mth UAV, there exists $L_m = \sum_{i\in\mathcal{A}_m'} \Gamma_{m,i}$ angle ranges. Since $\lambda_{k,m}{}^*(t)$ remains constant within each angle range, let $g_{m,i,j}(k) = \{0,1,2\}$ denote the communication scheduling of the kth node in $\mathcal{A}_m'$ within $\left[\psi_{m,i,(j-1)}, \psi_{m,i,j}\right)$, where the values of function $g_{m,i,j}(k)$ have the same meaning with the values of $\lambda_{k,m}(t)$. Then the variables with respect to t can be redefined as variables with respect to θ_m, i.e., $P_k'(\theta_m)$, $d_{k,m}'(\theta_m)$ and $h_{k,m}'(\theta_m)$, denoting the transmission power, the distance to the mth UAV, and the channel gain of the kth node in $\mathcal{A}_m'$, respectively. For simple presentation, setting $g_{m,i,j}{}^{-1}(1) = k^1_{m,i,j}$ and $g_{m,i,j}{}^{-1}(2) = k^2_{m,i,j}$, the achieved sum rate of the mth UAV at θ_m can be calculated as

$$R_m\left(\theta_m\right) = \log_2\left(\frac{\sigma^2 + P_{k^1_{m,i,j}}{}'\left(\theta_m\right) h_{k^1_{m,i,j},m}{}'\left(\theta_m\right)}{\sigma^2}\right) + \log_2\left(\frac{\sigma^2 + P_{k^2_{m,i,j}}{}'\left(\theta_m\right) h_{k^2_{m,i,j},m}{}'\left(\theta_m\right)}{\sigma^2 + P_{k^1_{m,i,j}}{}'\left(\theta_m\right) h_{k^1_{m,i,j},m}{}'\left(\theta_m\right)}\right). \tag{16}$$

The power constraint for SIC is updated as

$$P_{k^2_{m,i,j}}{}'\left(\theta_m\right) h_{k^2_{m,i,j},m}{}'\left(\theta_m\right) - P_{k^1_{m,i,j}}{}'\left(\theta_m\right) h_{k^1_{m,i,j},m}{}'\left(\theta_m\right) \geq P_{\text{tol}}. \tag{17}$$

Then the power control problem over time t can be rewritten as the problem over angle θ_m as follows

$$(\mathbf{P2.1}): \max_{P_k'(\theta_m)} R_m\left(\theta_m\right) \tag{18}$$

$$\text{s.t.} \quad 0 \leq P_k'\left(\theta_m\right) \leq P_{\max}, \ \forall k \in \mathcal{A}_m' \tag{18a}$$

(17).

Each problem (**P2.1**) over different θ_m is individual. As [21] has proved that the objective function (18) is concave, the Lagrange multiplier method and Karush–Kuhn–Tucker (KKT) conditions [44] can be applied to obtain the optimal solutions, given as

$$P_{k^1_{m,i,j}}{}^*\left(\theta_m\right) = \frac{P_{\max} h_{k^2_{m,i,j},m}{}'\left(\theta_m\right) - P_{\text{tol}}}{h_{k^1_{m,i,j},m}{}'\left(\theta_m\right)}, \tag{19a}$$

$$P_{k^2_{m,i,j}}{}^*\left(\theta_m\right) = P_{\max}. \tag{19b}$$

3.3. UAV trajectory optimization

According to the previous subsections, the optimal $\{\mathcal{A}_m\}$, $\Lambda^*(t)$, and $P_k{}^*\left(\theta_m\right)$ have been achieved. In this subsection, we first decompose the UAV trajectory into speed-free trajectory and maximum-speed trajectory, and then reformulate the original trajectory optimization problem as a speed-free trajectory optimization problem. Finally, this time-continuous problem is transformed into a conventional linear programming problem to be efficiently solved.

3.3.1. Trajectory decomposition

The UAV trajectory is first defined as follows.

- *Speed-constrained trajectory* $\{\theta_m(t)\}_{t=0}^{T}$: This corresponds to the UAVs' initial trajectories with the maximum angular velocity constraint $\left|\dot{\theta}_m(t)\right| \leq \omega^m_{\max}$.
- *Maximum-speed trajectory* $\{\bar{\theta}_m(t)\}_{t=0}^{\bar{T}_m}$: In this trajectory, the UAVs complete their circular trajectories at the maximum angular velocity with duration $\bar{T}_m = \frac{2\pi}{\omega^m_{\max}}$
- *Speed-free trajectory* $\{\tilde{\theta}_m(t)\}_{t=0}^{\tilde{T}_m}$: As discussed, the duration for speed-free trajectory is $\tilde{T}_m = T_m - \bar{T}_m$. In this trajectory, the UAVs can fly to arbitrary locations without speed constraint.

Based on the above definitions, the following lemma can be presented which follows from [34].

Lemma 2. *For any given speed-constrained circular trajectory* $\{\theta_m(t)\}_{t=0}^{T}$, *we can always construct a maximum-speed circular trajectory* $\{\bar{\theta}_m(t)\}_{t=0}^{\bar{T}_m}$ *and a speed-free trajectory* $\{\tilde{\theta}_m(t)\}_{t=0}^{\tilde{T}_m}$, *such that* $\{\theta_m(t)\}_{t=0}^{T} = \{\bar{\theta}_m(t)\}_{t=0}^{\bar{T}_m} \cup \{\tilde{\theta}_m(t)\}_{t=0}^{\tilde{T}_m}$. *Let* $C\left(\{\theta_m(t)\}\right)$ *denote the rate region achieved by the speed-constrained trajectory of the* m*th UAV, and* $\hat{C}\left(\{\bar{\theta}_m(t)\}, \{\tilde{\theta}_m(t)\}\right)$ *denote the rate region achieved by the combined trajectories. Then we have*

$$C\left(\{\theta_m(t)\}\right) = \hat{C}\left(\{\bar{\theta}_m(t)\}, \{\tilde{\theta}_m(t)\}\right). \tag{20}$$

3.3.2. Problem transformation and solutions

It is worth noting from Lemma 2 that the maximum-speed circular trajectory of the mth UAV is fixed, which can be given by

$$\bar{\theta}_m(t) = \omega^m_{\max} t, \quad \forall t \in \left[0, \bar{T}_m\right). \tag{21}$$

Then the average rate of the kth node with trajectory $\{\bar{\theta}_m(t)\}_{t=0}^{\bar{T}_m}$ can be calculated as

$$\bar{R}_{k,m}{}^* = \frac{1}{\bar{T}_m}\int_0^{\bar{T}_m} r_{k,m}(t)\mathrm{d}t \tag{22}$$

By substituting (21) into (22), we can apply the method of integration by substitution to convert $\bar{R}_{k,m}{}^*$ from a definite integral of t to a definite integral of θ_m (given in (23) inside Box I).

Although the integral function in formula (23) is difficult to get a closed-form solution, the constant solution of its definite integral can be easily obtained by the "*int*" function in Matlab. With the achieved $\bar{R}_{k,m}{}^*$, the speed-constrained trajectory optimization problem can be equivalent to a speed-free trajectory optimization problem, and can be relaxed as follows

$$(\mathbf{P2.2}): \max_{\tilde{\theta}_m(t)} \quad \tilde{R}_m \tag{24}$$

$$\text{s.t.} \quad \int_0^{\tilde{T}_m} r_{k,m}(t)\mathrm{d}t \geq \eta_c - \bar{T}_m \bar{R}_{k,m}{}^*, \forall k \in \mathcal{A}_m \tag{24a}$$

where $\tilde{R}_m$ is the average sum rate of the mth UAV with trajectory $\{\tilde{\theta}_m(t)\}_{t=0}^{\tilde{T}_m}$. Since the trajectory $\tilde{\theta}_m(t)$ is speed-free, it is independent of t and thus $\tilde{R}_m$ can be expressed as a function only related to θ_m and $\tilde{T}_m$. When the angle of the UAV is $\tilde{\theta}_{m,i,j} \in \left[\psi_{m,i,(j-1)}, \psi_{m,i,j}\right)$, the achievable rate of the kth node is given in (25).

$$\tilde{R}_{k,m}\left(\tilde{\theta}_{m,i,j}\right) = \log_2\left(\frac{\sigma^2 + \sum_{n=1}^{g_{m,i,j}(k)} P_{g^{-1}_{m,i,j}(n)}{}^*\left(\tilde{\theta}_{m,i,j}\right) h_{g^{-1}_{m,i,j}(n),m}\left(\tilde{\theta}_{m,i,j}\right)}{\sigma^2 + \sum_{n=1}^{g_{m,i,j}(k)-1} P_{g^{-1}_{m,i,j}(n)}{}^*\left(\tilde{\theta}_{m,i,j}\right) h_{g^{-1}_{m,i,j}(n),m}\left(\tilde{\theta}_{m,i,j}\right)}\right). \tag{25}$$

And the corresponding UAV's achievable rate is given in (26) inside Box II.

[4] For the IoT nodes, the communication scheduling results are equivalent to the user grouping results, by the definition of $\lambda_{k,m}(t)$. Once $\Lambda^*(t)$ is obtained, the random variable $h_{\pi^m_m(t),m}(t)$ can be subsequently handled as a constant only related to distance.

$$\bar{R}_{k,m}{}^{*} = \frac{1}{w_{\max}^{m}\bar{T}_{m}} \sum_{i=1}^{K_m} \sum_{j=1}^{\Gamma_{m,i}} \int_{\psi_{m,i,(j-1)}}^{\psi_{m,i,j}} \log_2 \left(\frac{\sigma^2 + \sum_{n=1}^{g_{m,i,j}(k)} P_{g_{m,i,j}^{-1}(n)}{}^{*}\left(\theta_m\right) h_{g_{m,i,j}{}^{-1}(n),m}\left(\theta_m\right)}{\sigma^2 + \sum_{n=1}^{g_{m,i,j}(k)-1} P_{g_{m,i,j}{}^{-1}(n)}{}^{*}\left(\theta_m\right) h_{g_{m,i,j}^{-1}(n),m}\left(\theta_m\right)} \right) \mathrm{d}\theta_m. \tag{23}$$

Box I.

$$\tilde{R}_m\left(\tilde{\theta}_{m,i,j}\right) = \log_2\left(1 + \frac{P_{k_{m,i,j}^{1}}{}^{*}\left(\tilde{\theta}_{m,i,j}\right) h_{k_{m,i,j}^{1},m}\left(\tilde{\theta}_{m,i,j}\right)}{\sigma^2}\right) + \log_2\left(1 + \frac{P_{k_{m,i,j}^{2}}{}^{*}\left(\tilde{\theta}_{m,i,j}\right) h_{k_{m,i,j}^{2},m}\left(\tilde{\theta}_{m,i,j}\right)}{\sigma^2 + P_{k_{m,i,j}^{1}}{}^{*}\left(\tilde{\theta}_{m,i,j}\right) h_{k_{m,i,j}^{1},m}\left(\tilde{\theta}_{m,i,j}\right)}\right). \tag{26}$$

Box II.

According to (25), the one-dimension searching can be adopted over the region $\left[\psi_{m,i,(j-1)}, \psi_{m,i,j}\right)$ to find the optimal solution $\tilde{\theta}_{m,i,j}{}^{*}$, which maximizes $\tilde{R}_m\left(\tilde{\theta}_{m,i,j}\right)$. When achieving the L_m optimal hovering locations $\left\{\tilde{\theta}_{m,i,j}{}^{*}\right\}$, denotes the UAV's hovering time at $\tilde{\theta}_{m,i,j}{}^{*}$ by $\tilde{T}_{m,i,j}$, then the problem (**P2.2**) can be transformed as follows

$$(\mathbf{P2.3}): \max_{\{\tilde{T}_{m,i,j}\}} \sum_{i=1}^{K_m} \sum_{j=1}^{\Gamma_{m,i}} \tilde{T}_{m,i,j} \tilde{R}_m\left(\tilde{\theta}_{m,i,j}{}^{*}\right) \tag{27}$$

$$\text{s.t.} \quad \sum_{i=1}^{K_m} \sum_{j=1}^{\Gamma_{m,i}} \tilde{T}_{m,i,j} \le \tilde{T}_m, \tag{27a}$$

$$\sum_{i=1}^{K_m} \sum_{j=1}^{\Gamma_{m,i}} \tilde{T}_{m,i,j} \tilde{R}_{k,m}\left(\tilde{\theta}_{m,i,j}{}^{*}\right) \ge \tilde{\eta}_c. \tag{27b}$$

where $\tilde{\eta}_c = \eta_c - \bar{T}_m \bar{R}_{k,m}{}^{*}$. Since $\tilde{\theta}_{m,i,j}{}^{*}$, $\tilde{R}_m\left(\tilde{\theta}_{m,i,j}{}^{*}\right)$, and $\tilde{R}_{k,m}\left(\tilde{\theta}_{m,i,j}{}^{*}\right)$ are constant values, problem (**P2.2**) is converted from a complicated time-continuous problem to a conventional L_m-dimensional linear programming problem. Thus, it can be efficiently solved by standard optimization techniques such as simplex method and existing software toolbox such as CVX [45]. Let $\left\{\tilde{T}_{m,i,j}{}^{*}\right\}$ denote the optimal solutions to problem (**P2.3**), and the corresponding optimal speed-free trajectory is $\tilde{\theta}_m{}^{*}(t)$. The optimal trajectory of the mth UAV can finally be achieved by combining $\tilde{\theta}_m{}^{*}(t)$ and $\bar{\theta}_m(t)$, and the floechart for trajectory design is given in Fig. 3. More precisely, the optimal trajectory $\theta_m{}^{*}(t)$ follows the successive hover-and-fly structure: The UAV anticlockwise flies from the initial angle $\theta_m(0) = 0$ to the final angle $\theta_m(T) = 2\pi$, during which it successively hovers above locations $\boldsymbol{q}_m\left(\tilde{\theta}_{m,1,1}{}^{*}\right), \ldots, \boldsymbol{q}_m\left(\tilde{\theta}_{m,K_m,\Gamma_{m,K_m}}{}^{*}\right)$, with durations $\tilde{T}_{m,1,1}{}^{*}, \ldots, \tilde{T}_{m,K_m,\Gamma_{m,K_m}}{}^{*}$, respectively, and flies among these locations at the maximum speed $V_{\max}$.

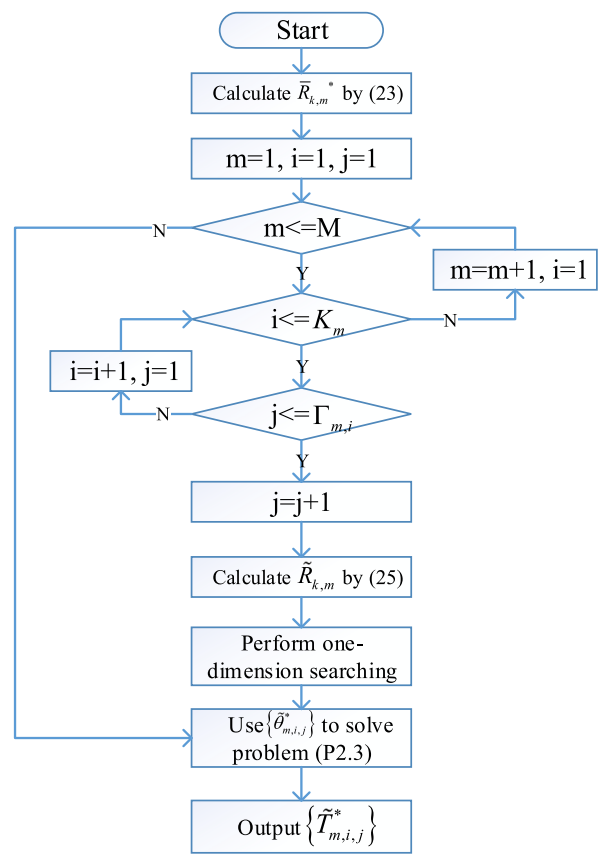

Fig. 3. The flowchart for UAV trajectory design.

3.4. Convergence and computational complexity analysis

The proposed algorithm to solve problem (**P1**) has been presented above, and the theoretical analysis of its convergence and computational complexity are analyzed as follows.

First, it should be noted that once the optimal $\Lambda^{*}(t)$ is obtained, it is independent of optimizing the transmission power and the flight trajectories of the UAVs. Since $\boldsymbol{p}^{*}(t)$ satisfies the KKT conditions and the trajectory design problem is finally transformed to a L_m-dimensional linear programming problem which can be solved by standard optimization techniques, the obtained $\boldsymbol{p}^{*}(t)$ and $\boldsymbol{q}_m{}^{*}(t)$ are global optimal solutions. Thus the overall algorithm does not require iteration, and the convergence analysis is only related to the clustering algorithm.

For the SCMM clustering algorithm, let C denote one match of graph G (the clustering results $\{\mathcal{A}_m\}$ can be achieved by C), and the matching weight is denoted by $W(C)$. The matching result will not be updated in the $(n+1)$th iteration unless it is higher than the current value. Therefore, we have

$$W(C^{(n+1)}) \ge W(C^{n}). \tag{28}$$

It should be noted that the matching weight is calculated based on the opposite of the Euclidean distance between UAVs and terrestrial nodes, and is thus upper-bounded by a finite value. Consequently, the non-decreasing sequence will converge to a finite value. Additionally, the complexity of the clustering algorithm is given by $\mathcal{O}\left(\left(K_{\max} M\right)^3\right)$, and for the simplified trajectory optimization problem, the complexity is $\mathcal{O}\left(K^2\right)$. To conclude, the total complexity is polynomial over the number of UAVs and terrestrial nodes.

Table 1
Simulation Parameters.

Parameter	Value	Parameter	Value
K	19	M	4
$K_{\max}$	5	H	100 m
γ_0	−45 dB	P_{tol}	−70 dBm
B	10 MHz	$V_{\max}$	90 m/s
$d_{\min}$	10 m	σ^2	−100 dBm
$\tilde{\alpha}$	2		

4. Numerical results

In this section, we present numerical results to validate the performance enhancement of our proposed joint optimization scheme for resource scheduling and trajectory design. The effectiveness of the node clustering and user grouping algorithms is presented, and the proposed trajectory optimization scheme is compared with two benchmark schemes in NOMA, FDMA and TDMA, respectively. The coordinates of the terrestrial nodes are randomly generated, and the other detailed simulation parameters are listed in Table 1.

Fig. 4 depicts different clustering results of the IoT nodes with four algorithms, seen in (a) Proposed SCMWM algorithm, (b) Clustering algorithm in [31], (c) Spectral clustering algorithm, and (d) Conventional mechanism [42]. For the algorithm in [31], the M points of Φ_C are first generated uniformly at the edge of the cell by [28, formula (5)], and then the nodes are clustered into M clusters based on Φ_C with the K-means clustering algorithm. As can be observed form Fig. 4, the distribution of nodes among the M clusters is more even with the proposed SCMWM algorithm. This is because the proposed algorithm can handle the maximum service capacity constraints of UAVs, thus ensuring the fairness among the nodes. In contrast, none of the remaining three algorithms can handle this constraint, and the simulation results confirm this fact. Specifically, the algorithm in [31] can only satisfy the uniform distribution to a certain extent by making Φ_C generated uniformly, while the spectral clustering algorithm and the conventional mechanism both have strong limitations, and make the load of each UAV quite imbalanced.

Since Fig. 4 has revealed the limited network capacity of the spectral clustering algorithm and conventional mechanism, they are not discussed any more in the following. Fig. 5 further compares capacity of the NOMA network between the proposed SCMWM algorithm and the algorithm in [31], with maximum transmit power $P_{\max} = 20$ dBm and flight period $T = 20$ s/22 s/24 s. It can first be observed that when the upload requirement of each node increases, the average sum rate with both clustering algorithms will decrease gradually, and the benefit of the proposed SCMWM algorithm is obvious with all flight periods. Moreover, the decline curves with the algorithm in [31] tend to be more rapid with increasing η_c, bringing larger performance gap between the two clustering algorithms. It should also be noted that when $T = 20$ s/22 s and $\eta_c > 0.3$ Gbits/0.34 Gbits, there will be no feasible solutions to problem (**P**2) with the algorithm in [31], meaning that the resources in the UAV-assisted subnetwork can no longer meet the upload requirements now. The above simulation results indicate that the flight period should be adjusted properly according to different upload requirements. Furthermore, they present the significance of the clustering results in improving the network capacity and the advantage of our proposed SCMWM algorithm, which is meaningful in scenarios with a large number of distributed users.

Based on the proposed SCMWM algorithm, the detailed grouping and communication scheduling results for UAVs are first presented in Table 2. Within each angle range $[\psi_{m,i,(j-1)}, \psi_{m,i,j})$, two selected nodes are formed into one group to communicate with the UAV via NOMA, and the user 2 has a better channel gain which is decoded first. Then, Fig. 6 depicts the optimized UAV trajectories in NOMA network with node upload requirements of (a) $\eta_c = 0.1$ Gbits, (b) $\eta_c = 0.2$ G bits, and (c) $\eta_c = 0.3$ Gbits, where the flight period is set to be 20 s and $P_{\max} = 20$ dBm. It can be observed that as η_c increases, the hovering positions of each UAV increases correspondingly. The hovering positions in scenario (a) are the global optimal locations of the UAVs, where $\bar{R}_{k,m}$ with the maximum-speed trajectory has satisfied the node upload requirements and the UAV can hover at the global optimal location during the remaining time. However, as η_c increases, UAVs should exploit sub-optimal locations to satisfy each node's requirement, while sacrifice some network performance. Furthermore, the optimized UAV trajectories in TDMA network are given in Fig. 7. It can be seen that the UAVs need more hovering locations in TDMA, which

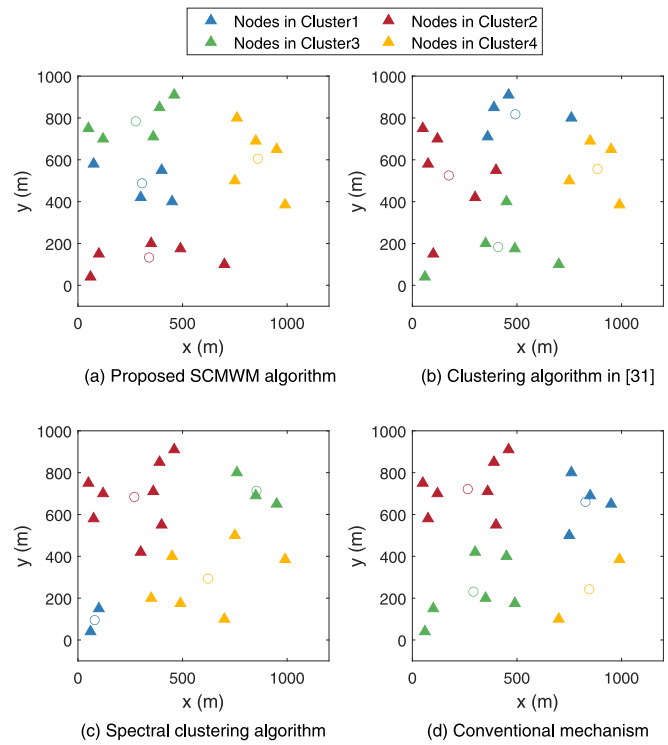

Fig. 4. Clustering results comparison among four algorithms.

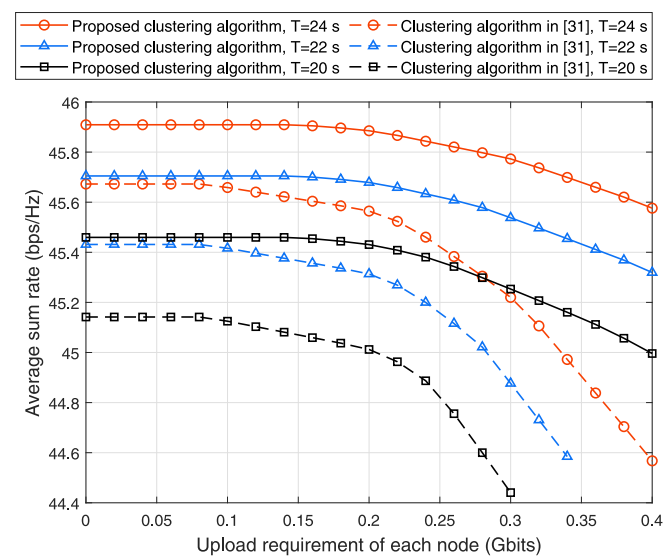

Fig. 5. Average sum rate of the network for different clustering algorithms versus node upload requirements.

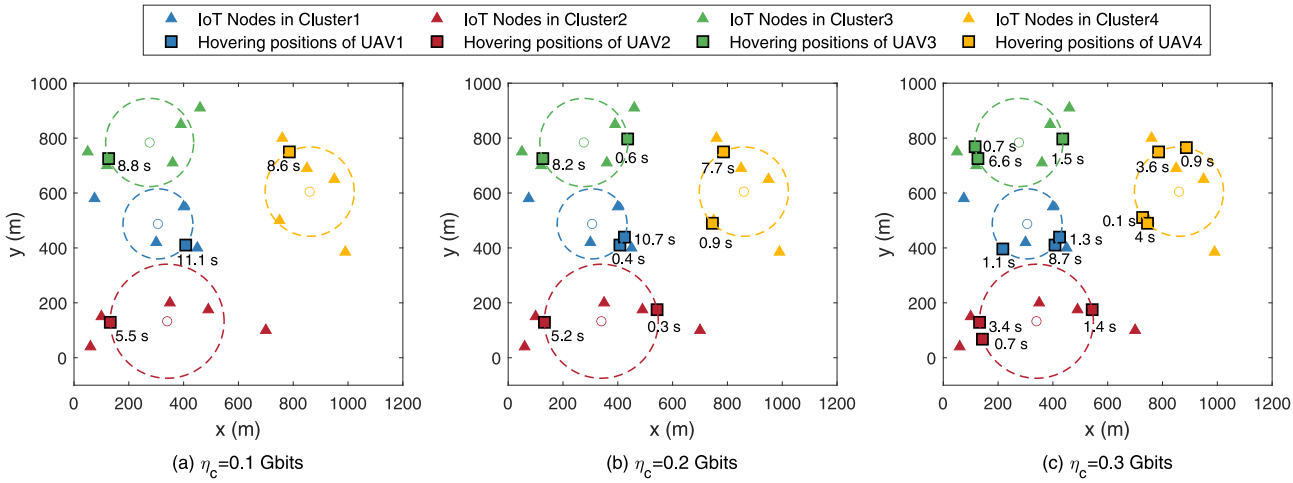

Fig. 6. Optimized UAV trajectories versus different node upload requirements with randomly distributed nodes in NOMA network.

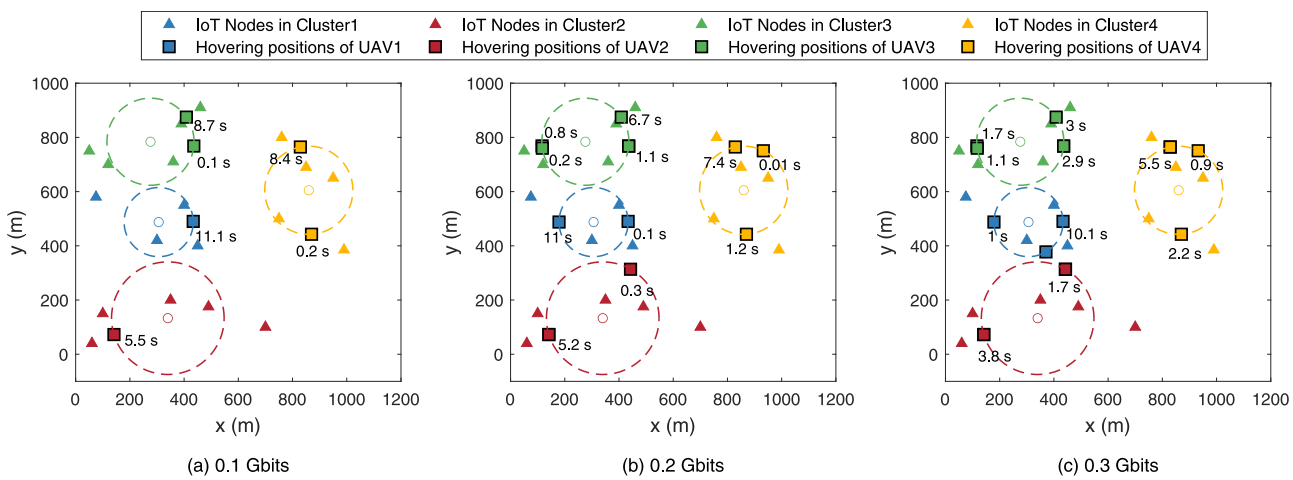

Fig. 7. Optimized UAV trajectories versus different node upload requirements with randomly distributed nodes in TDMA network.

Table 2
The grouping and communication scheduling results for UAVs.

(a) UAV 1

Angle range	[0.02, 1.45)	[1.45, 2.11)	[2.11, 2.29)	[2.29, 3.14)	[3.14, 3.94)	[3.94, 5.24)	[5.24, 5.82)	[5.82, 6.30)
User 1	Node 4	Node 2	Node 1	Node 3	Node 2	Node 4	Node 3	Node 1
User 2	Node 1	Node 1	Node 2	Node 2	Node 3	Node 3	Node 4	Node 4

(b) UAV 2

Angle range	[0.09, 0.31)	[0.31, 1.05)	[1.05, 1.84)	[1.84, 2.30)	[2.30, 2.72)	[2.72, 3.44)	[3.44, 4.92)	[4.92, 4.97)	[4.97, 6.37)
User 1	Node 5	Node 2	Node 1	Node 3	Node 2	Node 4	Node 3	Node 5	Node 5
User 2	Node 1	Node 1	Node 2	Node 2	Node 3	Node 3	Node 4	Node 4	Node 1

(c) UAV 3

Angle range	[−0.10, 0.08)	[0.08, 1.96)	[1.96, 2.17)	[2.17, 3.14)	[3.14, 3.24)	[3.24, 4.18)	[4.18, 4.51)	[4.51, 5.08)	[5.08, 6.18)
User 1	Node 5	Node 2	Node 2	Node 2	Node 4	Node 3	Node 5	Node 4	Node 1
User 2	Node 1	Node 1	Node 1	Node 3	Node 3	Node 4	Node 4	Node 5	Node 5

(d) UAV 4

Angle range	[−0.28, 1.11)	[1.11, 1.4)	[1.40, 1.76)	[1.76, 2.85)	[2.85, 3.76)	[3.76, 4.77)	[4.77, 5.20)	[5.20, 6.00)
User 1	Node 5	Node 1	Node 3	Node 2	Node 3	Node 5	Node 4	Node 1
User 2	Node 1	Node 2	Node 2	Node 3	Node 4	Node 4	Node 5	Node 5

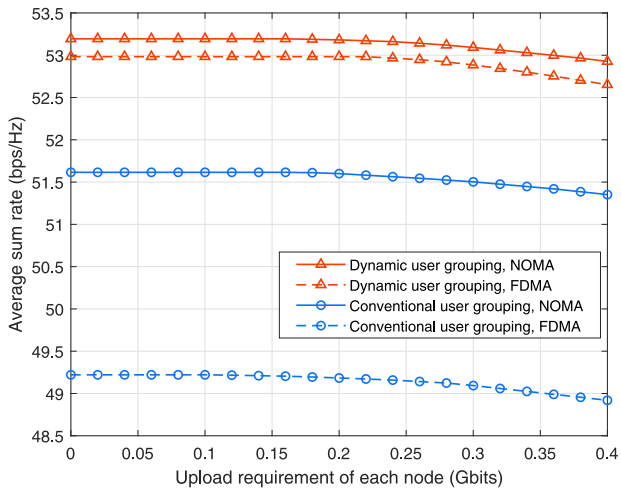

Fig. 8. Average sum rate of the network versus node upload requirements.

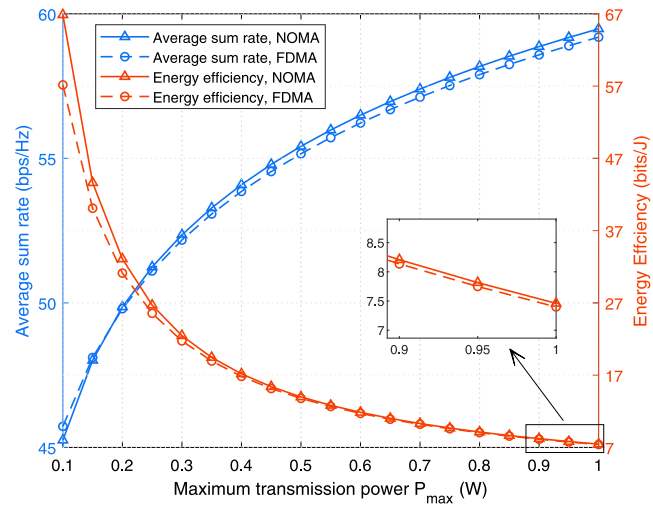

Fig. 9. The performance gap between NOMA and FDMA versus $P_{\max}$ in terms of average sum rate and energy efficiency.

correspondingly decrease their performance in communication capacity as shown in Fig. 10.

Fig. 8 analyzes the effect of user grouping scheme on the average sum rate by presenting the performance differences between two different user grouping schemes in NOMA and FDMA, with $T = 24$ s and $P_{\max} = 25$ dBm. For widely adopted conventional grouping schemes in NOMA, the users who experience significantly different channel coefficients are grouped together [46]. This grouping scheme is to guarantee the fairness in each group and maximize the performance gap between NOMA and OMA, and this theory is consistent with Fig. 8. In contrast, our dynamic user grouping scheme pairs two highest channel gain nodes into a group, due to the reason that this scheme is time-varying and all the nodes can experience a high channel gain in turns. Although such a grouping scheme reduces the performance gap between NOMA and FDMA, it improves the performance of both access protocols, and the proposed grouping scheme has a significantly higher average sum rate than the conventional scheme in both access scenarios, thus proving its effectiveness.

Fig. 9 further shows the performance gap between NOMA and FDMA in terms of average sum rate and energy efficiency versus the maximum transmission power, with $T = 20$ s and $\eta_c = 0.3$ Gbits. First, when $P_{\max} < 0.2$ W, the average sum rate in FDMA is slightly higher than in NOMA. This is due to the fact that in our proposed NOMA grouping scheme where the user channel gain difference is relatively small, the transmission power of the weaker channel gain user will thus be severely restricted with small $P_{\max}$, which will result in a rate degradation. Consequently, the sum rate in NOMA increases with increasing $P_{\max}$, as well as the performance gap between NOMA and FDMA. As for energy efficiency, we can see that the NOMA network maintains better performance than FDMA network all the time. When $P_{\max} = 0.1\ W$, the NOMA network can improve the energy efficiency performance by about 17% than FDMA network, but the performance gap decreases gradually. The reason is that when $P_{\max}$ increases, the power restriction in NOMA network is released and both nodes transmit their signals at the maximum power, which reduces their energy efficiency performance gap.

Finally, to validate the performance enhancement of our proposed trajectory design, we compare it with the following two benchmark schemes.

- *Successive user visiting and hovering (UVH)*: The mth UAV visits the nodes in $\mathcal{A}'_m$ in turns at the maximum speed $V_{\max}$, and hovers at the optimum locations among the nodes. The hovering locations, hovering duration and the resource scheduling are jointly optimized, where the resource scheduling means user pairing, decoding order, and power control in NOMA, and user pairing and communication time allocation in FDMA and TDMA.
- *Static hovering*: The M UAVs are deployed to hover over their optimized locations as in [31] throughout the flight period, and the optimization problem corresponds to a conventional communication resource scheduling problem in NOMA, FDMA, and TDMA, respectively.

Fig. 10 compares the average sum rate of our trajectory design with the two benchmark schemes among different access protocols, with $P_{\max} = 30$ dBm and $\eta_c = 0.25$ Gbits. It can first be seen that the extended flight period T results in the performance enhancement for all trajectory schemes under consideration, with a gradually flat trend. Furthermore, our proposed trajectory design significantly outperforms the other two trajectory schemes, which reveals the importance of utilizing UAVs' flexibility in improving the communication performance. Obviously, the static hovering scheme always performs the worst, because it hovers over one single location without movement all the time. It is also worthy noting that when $T = 15$ s, there will be no feasible solutions for the successive UVH trajectory, and the main reason is that the flight distance with this trajectory is the longest and more flight period is needed. For access protocol comparison, both NOMA and FDMA achieve higher rates than TDMA with all trajectory schemes, while NOMA realizes a higher average sum rate than FDMA with the proposed trajectory, but performs slightly worse than FDMA with the successive UVH trajectory. This is due to the fact that when the UAV flies with the successive UVH trajectory, the grouped users at time instant t are nodes at both ends of the UAV's flight line this time. Therefore, the channel differences are smaller than the proposed trajectory, and the restricted power control of the weaker channel gain user results in a slight rate degradation. However, the energy efficiency in NOMA is still better than that in FDMA with this case, as seen in Fig. 9.

5. Conclusion

In this paper, we have studied the joint resource scheduling and trajectory design optimization in a multi-UAV-assisted uplink NOMA network. Specifically, to handle the challenging problem which contains integral variables and highly coupled non-convex objective functions and constraints, we decompose the original problem into three sub-problems, namely nodes clustering and dynamic user grouping, transmission power control, and UAV trajectory design. To solve each

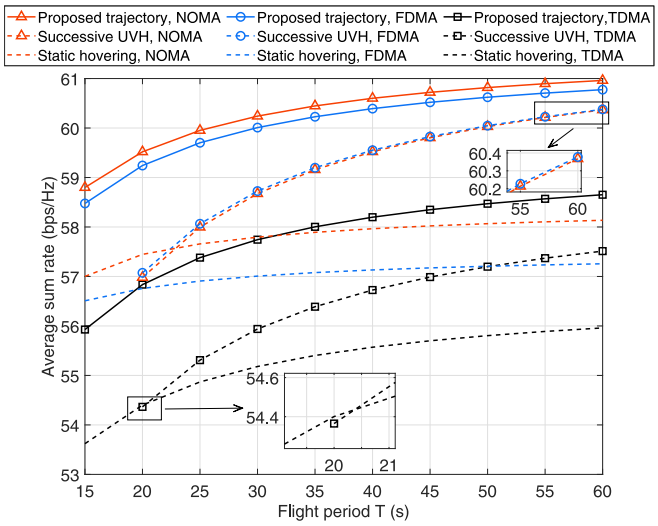

Fig. 10. The average sum rate based performance comparison versus T with NOMA, FDMA, and TDMA protocols.

sub-problem, the corresponding algorithms are described, as well as the principles for optimizing the variables. Finally, numerical results validate the superiority of the joint optimization scheme, and presents its improvement on the average sum rate and energy efficiency, which will make sense for resource scheduling designs of the future network.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgment

The work was supported by the National Natural Science Foundation of China under Grant 61901509.

Appendix. Proof of Lemma 1

To start with, we denote the distance between the UAV and the ith node in $\mathcal{A}_m{}'$ by $d_{m,i}{}'(t)$. Based on the definition of $\Psi_{m,i}$, we have $d_{m,(i+1)}{}'(t) \le d_{m,(i+2)}{}'(t)$ when $\Psi_{m,(i-1)} \le \theta_m(t) < \Psi_{m,(i+1)}$, and $d_{m,i}{}'(t) \le d_{m,(i+1)}{}'(t)$ when $\theta_m(t) \le \Psi_{m,i}$. Since $\Psi_{m,(i+1)} < \Psi_{m,i}$, it can be easily derived that

$$d_{m,i}{}'(t) \le d_{m,(i+1)}{}'(t) \le d_{m,(i+2)}{}'(t),\ \Psi_{m,(i-1)} \le \theta_m(t) < \Psi_{m,(i+1)}. \tag{29}$$

Similarly, we have $d_{m,(i+1)}{}'(t) \le d_{m,i}(t)$ when $\Psi_{m,i} \le \theta_m(t) < \Psi_{m,(i+2)}$, and $d_{m,(i+2)}{}'(t) \le d_{m,(i+1)}{}'(t)$ when $\theta_m(t) \ge \Psi_{m,(i+1)}$. Thus the similar result can be obtained as

$$d_{m,(i+2)}{}'(t) \le d_{m,(i+1)}{}'(t) \le d_{m,i}{}'(t), \quad \Psi_{m,i} \le \theta_m(t) < \Psi_{m,(i+2)}. \tag{30}$$

The distance relations in (29) and (30) complete the proof for Lemma 1. ■

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